Best Practices for Biomedical Research Data Management

Module 1: Introduction and Overview Data Management Services

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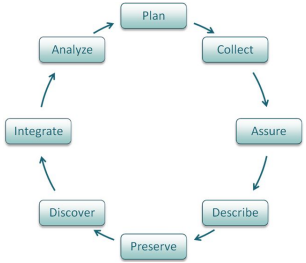

This video will walk through an example of a librarian working with researchers to create a data management plan and cover some of the parts that go into a DMP. It can help researchers understand the support role an information professional (librarian) can play in research data management planning as well as illustrate one approach to constructing concise DMPs.

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Consulting on Data Management Plans

How does this work?

- Grants administrators contact someone in the library who is available to consult on DMP development with **lists** of faculty who are preparing NSF/NIH and other organization grant proposals

Consulting on Data Management Plans

List of faculty preparing grant proposals

Number	Proposals in Process	Funder	Type	Due Date	Project Title	Start Date	End Date	Department	Research Admin.	Status	Comments
Number	Researcher's Name	NSF	Transfer	N/A	Mathcamp	7/1/2011	6/30/2013	Mathematics		Submitted	
13		NSF	TUES	5/28/2012	Transforming Undergraduate Education in Science	N/A	N/A	Education		Not submitted	
14		NSF - Supplement	PI Transfer	5/31/2012	Research in Pairs at Canada/US Mathcamp	6/1/2012	12/31/2012	Mathematics		Submitted	
16		NSF GRFP	Fellowship	6/1/2012	GRE NSF Fellowship	6/1/2012	5/31/2015	Psychology		Submitted	
17		NSF GRFP	Fellowship	6/1/2012	GRFP NSF Fellowship	6/1/2012	5/31/2015	Electrical/Computer Enineering		Submitted	
18		NSF GRFP	Fellowship	6/1/2012	GRFP NSF Fellowship	6/1/2012	5/31/2015	Electrical/Computer Enineering		Submitted	
19		NSF GRFP	Fellowship	6/1/2012	GRFP NSF Fellowship	6/1/2012	5/31/2015	Biomedical Engineering		Submitted	
22		NSF	RTG	6/5/2012	Research Training Group in the Mathematical Sciences	N/A	N/A	Mathematics		Not submitted	
29		NSF	BIGDATA	6/13/2012	N/A	N/A	N/A	Classics		Not submitted	
36		NSF11-690	CAREER	7/1/2012	Faculty Early Career Development	N/A	N/A	Education		Not Submitted	
46		NSF		7/6/2012	Rapid Identification of Cryptotephras Using Magnetic Properties and Reflectance Spectroscopy	1/1/2013	12/31/2015	Geology		Submitted	
49		NSF 12-529	S-STEM	7/14/2012	Scholarships in Science, Technology, Engineering, and Mathematics (S-STEM)	N/A	N/A	Biology		Not Submitted	
51		NSF 12-552	REESE	7/17/2012	Research and Evaluation on Education in Science and Engineering (REESE)	7/1/2013	6/30/2016	Education		Submitted	
57		NSF	Career Bio	7/23/2012	Bicyclic peptides as conformational mimetics of phosphotyrosine	2/1/2013	1/31/2018	Chemistry		Submitted	
58		NSF	11-690	7/23/2012	Faculty Early Career Development	7/1/2013	6/30/2018	Physics & Astronomy		Submitted	
59		NSF CARRER	11-690	7/23/2012	Faculty Early Career Development	7/1/2013	6/30/2018	Biology		Submitted	

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- Use an **email template** to contact researchers

Data Management Plan Email Template

Below please find the information you'll need to begin developing your proposal's DMP.

*If I'm working with a faculty/researcher who I don't know I will often write a statement like this: "My name is *******, the Data Management Services Coordinator at ****** and I assist faculty with writing data management plans (DMP) for NSF grant submissions. ********, emailed me about your forthcoming NSF grant.

Please send me your grant summary and prospectus, this will help me understand your research proposal, as well as the type of data and metadata you will most likely be collecting.

I don't have the name/number of the NSF grant, please send this to me too.

I've attached a template for a generic NSF data management plan, meaning, if the grant's directorate has specific or additional requirements, we can add these to this template. If they don't refer to anything other than the NSF's general data management requirements, then this template is fine to use.

So to get you started with creating the plan take a look at our research guide we created on pulling together a data management plan **[insert link to library research/LibGuide or another institutional resource here.]**

Data Management Plan Email Template Continued

A few suggestions for moving ahead with writing the plan

- Refer to other parts of your proposal when you can.
- The other thing that often trips people up is the **metadata** piece – if someone opened up an Excel file of your data how would they know what they're looking at. What conventions of the field are you using in thinking about how you organize and name the columns of data?
- Is there a data dictionary or index that accompanies the Excel file to make sense of it all?

You'll have a stronger DMP if you think about all of those products of the research early on in the process and think about how you will want to clean them up/maintain /preserve/make available.

The other thing I can help with is in providing some **[insert institution's name here]** approved language on how often the servers get backed up (if you decide to store your data locally or in **[insert institution's name here]** Research Storage and **[insert institution's name here]** Acceptable Use and Intellectual Property policies.

Use the template to begin outlining your data management plan draft and then email it to me when finished; I will review it and then we can either meet in person to go over it or consult via email on the draft.

Please don't hesitate to contact me if you have any questions. Looking forward to working with you.

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- Will work with researcher in person, email, both
- Will arrange a meeting with subject specialist/metadata librarian (when necessary)

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- **DMP template** attached to email

Data Management Template

- NSF has published a revised version of the NSF Proposal and Award Policies and Procedures Guide (PAPPG) (NSF 11-1) that will require, in all proposals submitted, or due, on or after January 18, 2011, a supplementary document of no more than two pages describing a Data Management Plan (DMP) for the proposed research.

- In general, a Data Management Plan will describe the nature of the data or products generated by the research, and how these will be managed with regard to access, sharing and archiving.

There are the five (5) requirements to be addressed in the DMP by the revised PAPPG:

1. **Research Products**
2. **Data Format**
3. **Access to Data and Data Sharing Policies**
4. **Re-Use, Re-Distribution and Production of Derivatives**
5. **Archiving Data**

Data Management Template

1. Research Products

a. **Describe the types of data and products that will be generated in the research:**

- numerical data
- data systems such as spectra, diffraction patterns, physical properties, time- dependent information, processes, theoretical formalisms, and computational strategies
- final or intermediate numerical results from theoretical calculations, software, and curriculum materials

Data Management Template

2. Data Format

a. **Describe the format in which the data or products are stored**
   - hardcopy notebook and/or instrument outputs
   - ASCII, html, xml, jpeg or other formats

b. **If the data are stored in unusual formats**, explain how the data may be converted to a more accessible format or made available to interested parties

Data Management Template

3. Access to Data and Data Sharing Policies

a. **Describe your plans, if any, for providing such general access to data,** including websites maintained by your research group, and direct contributions to public databases

b. **Format of data sent to supplementary materials section of peer reviewed journals:** tables, graphs, computer codes

c. **What policies/practices do you follow in releasing data for access:** is the data posted before or after formal publication?

d. **Also include whether or not your data will be placed into databases that mine the published literature**
   - CAIDA, PubChem, Open Access Directory, Science Commons

e. **Describe policies for data sharing,** including where applicable, provisions for protection of privacy, confidentiality, intellectual property, national security, or other rights or requirements

Data Management Template

4. Re-Use, Re-Distribution and Production of Derivatives

a. **Describe your policies regarding the use of data provided via general access or sharing:** If you plan to provide data and images on your website, will the website contain disclaimers, or conditions regarding the use of the data in other publications or products?

b. **Describe these disclaimers and/or terms of use**

Data Management Template

5. Archiving Data

a. **Describe how data will be archived and how preservation of access will be handled**

b. **Is there a plan to transfer digitized information to new storage media or devices as technological standards or practices change?**

c. **Will there be an easily accessible index that documents where all archived data are stored and how they can be accessed?**

Consulting on Data Management Plans

Justification for not including a DMP:

A valid Data Management Plan may include only the statement that no detailed plan is needed, as long as the statement is accompanied by a clear justification. Proposers who feel that the plan cannot fit within the supplement limit of two pages may use part of the 15-page Project Description for additional data management information. Proposers are advised that the Data Management Plan may not be used to circumvent the 15-page Project Description limitation. The Data Management Plan will be reviewed as an integral part of the proposal, coming under Intellectual Merit or Broader Impacts or both, as appropriate for the scientific community of relevance.

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- Ask the researcher to provide grant summary and/or prospectus

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- Within response email will provide a ‘path’ or ‘outline’ for the researcher on how to proceed

Data Management Plans

Case Study: The "Tea Grant"

- Several on-campus and off-campus collaborators
- Data generated by several researchers and varied formats
- Research Products:
  - climate variability
  - secondary metabolite chemistry of tea leaves from *in situ* stratified sampling
  - manipulative experiments
  - farmer and consumer surveys
- IRB requirements

Data Management Plan Example

1. Research Products, Data Format and Data Sharing:

This project will generate research products on climate variability, secondary metabolite chemistry of tea leaves from *in situ* stratified sampling and manipulative experiments, and from farmer and consumer surveys. The specific products, format of data storage, and format of data sharing if listed below. All the project data will be stored using **** University Information Technology (UIT) resources. A main study deliverable is a web accessible catalogue of all data products generated from this project posted on a Spark 2.0 wiki infrastructure maintained by **** UIT. UIT offers a suite of Web 2.0 tools for communication and collaboration across **** University entitled Spark: Spark 2.0 tools include Wikis, Blogs, Podcast Publishing, Forums, Media Markups for web based video annotation, Google Map collections, and Web Meetings **(link to institutional URL).** The Spark wiki site will be made public to allow access to non**** users and the link to this website will be posted prominently on the websites of all project senior personnel. The web accessible data sharing platform will include an easily accessible index listing where the data are stored and how they can be accessed. This index will be stored electronically with the actual files. We will additionally share data through publication in open access journals where suitable and presentations at conferences, workshops, seminars and public lectures.

The farmer and consumer data products consist of human subjects research and (are) subject to IRB and privacy requirements. The raw data cannot be made publicly available. However the results will be published in peer reviewed journals and presented.

Data Management Plan Example

RESEARCH PRODUCTS	FORMAT OF DATA STORAGE	FORMAT OF DATA SHARING
A. Climate Variability		
Trends in precipitation and circulation over the study site	Atmospheric fields will be stored in netcdf format, one of the standard formats typically used in the ocean and atmospheric sciences	Netcdf format allows 4dimensional gridded data to be stored with all of the metadata; Web accessible catalogue
Statistical output from a classification and regression tree (CART) model examining the variability of climate and tea prices	Excel spreadsheets; SPSS software output	Excel spreadsheets on webaccessible catalogue; Writeup and graphs in scientific articles
Maps of climate trends	Digital .jpeg image files	Digital .jpeg image files on Web accessible catalogue
B. Secondary Metabolites of Leaves from *In Situ* Stratified Sampling and Manipulative Experiments		
Chromatogram and peak quantification output from secondary metabolite analysis	EMPOWER, LCLab and ChemStation software data analysis output; Excel spreadsheets	Chemical library maintained on webaccessible catalogue; Writeup and graphs in scientific articles
Laboratory Notes	Hardcopy notebooks; Digital copy of scanned notebook pages	Digital .jpeg files of scanned notebook pages on web accessible catalogue

Data Management Plan Example

NSF DMP Template: Describe your policies regarding the use of data provided via general access or sharing

RESEARCH PRODUCTS	FORMAT OF DATA STORAGE	FORMAT OF DATA SHARING
C. Farmer and Consumer Surveys		
Participant observation field notes	Digital copy of scanned notebook pages; Typed notes in word processing files	Collection notebooks and all datasets will be provided to ********* Institution as part of their efforts to gather legacy data related to ethnobiology and ethnoecology.
Questionnaire responses	Digital voice recordings; Typed notes in word processing files; Excel spreadsheets; Anthropac software data analysis output; SPSS software output	Ethnographic narrative on webaccessible catalogue; writeup and graphs in scientific articles
Social network data	Typed notes in word processing files; Excel spreadsheets; UCInet software data analysis output; .jpeg images	Digital .jpeg images on webaccessible catalogue; Writeup and graphs in scientific articles
Voucher specimens	Voucher specimens with typed descriptions using botanical standards distributed to **(various institutions)**; Scanned voucher specimens	Digital .jpeg images on web accessible catalogue international biodiversity informatics websites
Prior informed consent forms	Forms will be scanned	Not publically available as subject to privacy requirements

Data Management Plan Example

2. ReUse, ReDistribution and Production of Derivatives:

The PI will retain principal legal rights to the intellectual property developed under this grant as is compliant with the **** University's Policy on Rights and Responsibilities with Respect to Intellectual Property. Others, such as academic audiences, public policy practitioners, and local and state government are welcome to reuse or redistribute our data with the only caveat that they identify us as the source. Data will also be made available on distribution media through requests made to PI *****. Minor handling fees will be charged as needed for such requests to support the costs of media and postage. PI ***** will request appropriate acknowledgment in all publications or other use of data.

Data Management Plan Example

3. Archiving Data:

All the collected data will be archived on the **** University Information Technology (UIT) research storage that is provisioned on **** NetApp enterprise networked storage. Data security and confidentiality are protected by using Microsoft Active Directory authentication, and the storage is backed up to LTO4 tape on a daily and weekly basis and stored offsite at Iron Mountain facilities. As storage needs for the research study increase over time, additional storage can be easily provisioned on the same appliance. UIT issues storage space in 50GB increments, up to several TB. Only the latest data (I year) will be made available online through the project website that will contain an easily accessible index to navigate the data and listing how they can be accessed while older data archived in the UIT research storage will be made available online for downloading in manageable sizes (l0GB) upon request. Updating the data on the website is expected to be done every two weeks.